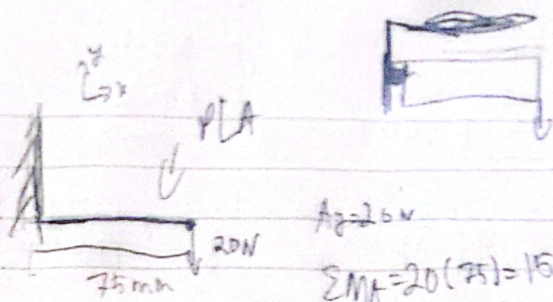




$$A_2 = 20 \text{ N}$$

$$\Sigma M_A = 20(75) = 1500 \text{ N}\cdot\text{mm}$$

$$I = \frac{bh^3}{12} \quad \sigma_{\max} = \frac{M_c}{I} \quad c = \frac{h}{2} \quad \sigma_{\max} = \frac{M(h/2)}{bh^3/12} \Rightarrow \frac{6M}{bh^2}$$

$$SF = \frac{6}{3}$$

$$\sigma_y = 40 \text{ MPa} \rightarrow \frac{40}{3} = 13.33 \text{ MPa}$$

$$PLA = 1155 \text{ MPa}$$

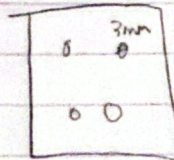
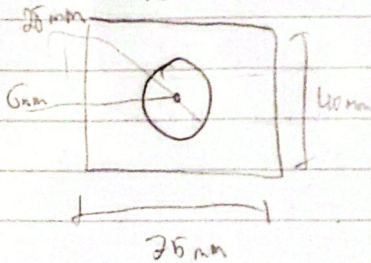
$$h_{\min} = \sqrt{\frac{6FL}{b\sigma_{\max}}} \rightarrow \sqrt{\frac{6(20)(75)}{(40)(13.33)}} \rightarrow 4.11 \text{ mm} \quad h \geq 4.11 \text{ mm} \text{ yield strength}$$

$$\delta_{\max} = \frac{FL^3}{3EI} \quad I = \frac{bh^3}{12} \quad \delta = \frac{FL^3}{3E(bh^3/12)} \rightarrow \frac{4FL^3}{Ebh^3}$$

$$h_{\min} \delta = \sqrt[3]{\frac{4FL^3}{E\sigma_{\max}}} \rightarrow 13.33 \text{ mm} \approx 15 \text{ mm} \text{ deflection}$$

$$I = \frac{(40)(15)^3}{12} = 11250 \text{ mm}^4$$

$$\delta = \frac{(20)(75)^3}{3(1188)(11250)} = 0.211 \text{ mm} \quad L = 30 \text{ mm}$$



$$A = \frac{\pi(3)^2}{4} = 7.07 \text{ mm}^2$$

$$\tau = \frac{F}{A} = \frac{15}{7.07} = 2.12 \text{ MPa}$$

4 x 3 mm bolts

